

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

THE VARIATIONAL AUTOENCODER

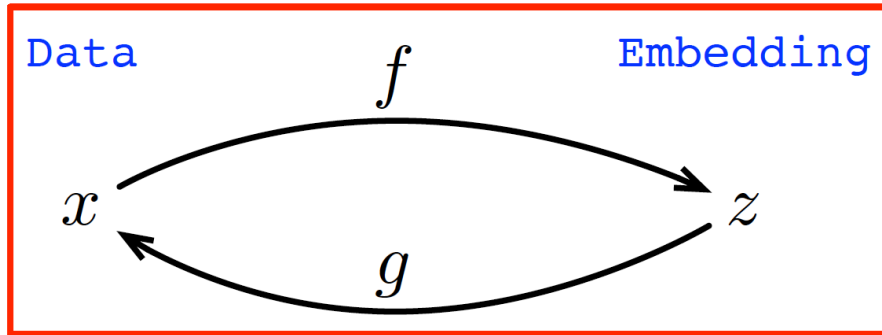
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Probabilistic Autoencoders



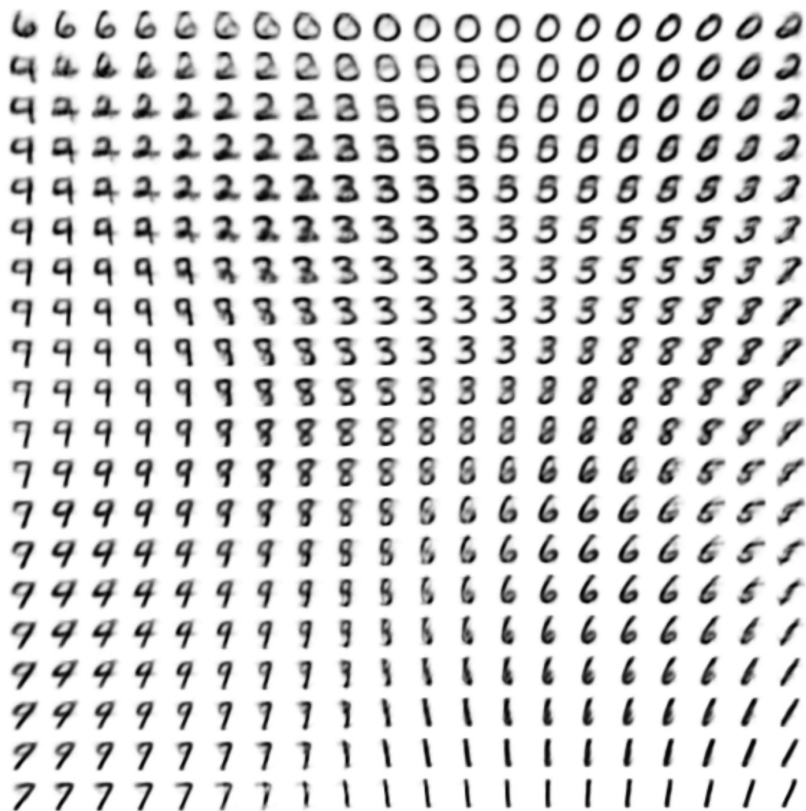
- $\mathcal{X}$: data space
- $\mathcal{Z}$: latent (embedding)

- $f(x)$ gives a **distribution over** $\mathcal{Z}$, denoted $q_{f(x)}$
- $g(z)$ gives a **distribution over** $\mathcal{X}$, denoted $p_{g(z)}$

Example: Binarized MNIST

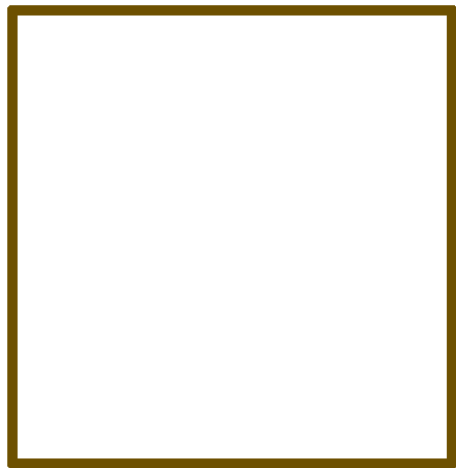
Kingma and Welling (2014). Autoencoding variational Bayes.

- $\mathcal{X} = \{0,1\}^{784}$
- $\mathcal{Z} = \mathbb{R}^2$

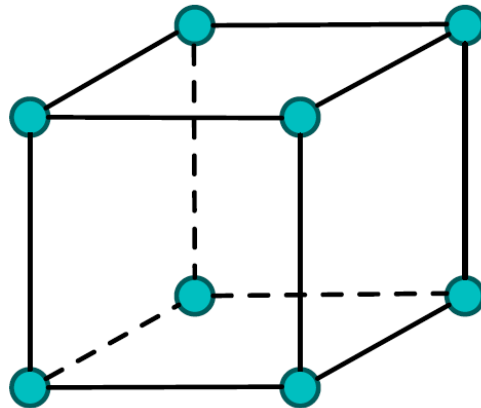


Binarized MNIST: the Mappings

$$\mathcal{Z} = \mathbb{R}^k$$



$$\mathcal{X} = \{0,1\}^{784}$$



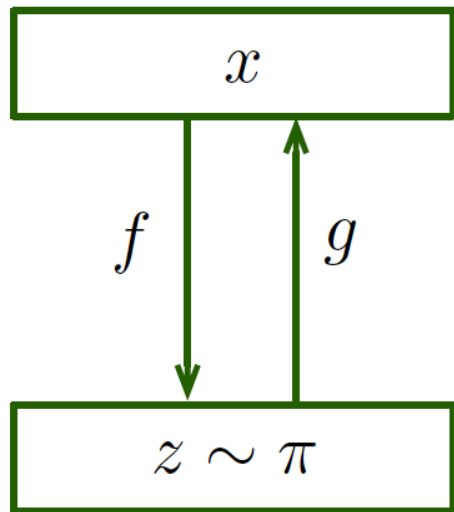
- $g(z)$ is a product-Bernoulli distribution over $\{0,1\}^{784}$
- $f(x)$ is a diagonal Gaussian distribution over $\mathbb{R}^k$

Binarized MNIST: the Architecture

- $g(z)$ is a product-Bernoulli distribution over $\{0,1\}^{784}$, i.e., $g: \mathbb{R}^k \rightarrow [0,1]^{784}$
- $f(x)$ is a diagonal Gaussian distribution over $\mathbb{R}^k$, i.e., $f: \{0,1\}^{784} \rightarrow \mathbb{R}^{2k}$

The Learning Process

Given data $x_1, \dots, x_n \in \mathcal{X}$, learn a model (f, g)



- $f(x)$ gives a distribution on $\mathcal{Z}$, denoted $q_{f(x)}$
- $g(z)$ gives a distribution on $\mathcal{X}$, denoted $p_{g(z)}$
- Prior π on $\mathcal{Z}$
- $\Pr(x, z|g) = \pi(z)p_{g(z)}(x)$
- Want $q_{f(x)}(z) \approx \Pr(z|x, g)$

Learning task: Choose f, g to maximize

$$\sum_{i=1}^n L_{f,g}(x_i), \text{ where } L_{f,g}(x) = \ln \Pr(x|g) - K(q_{f(x)}, \Pr(\cdot | x, g)).$$

Some More Pictures



0 6 7 7 8 1 4 8 2 8
9 6 8 3 4 6 0 3 1 4
3 3 7 1 3 0 9 1 7 9
8 9 0 8 6 4 1 4 6 3
8 2 3 3 3 3 1 3 8 6
6 4 4 8 6 1 6 6 6 6
4 5 2 6 6 5 1 8 4 9
9 9 8 7 3 7 2 8 2 3
0 4 6 1 2 3 2 0 8 8
9 7 5 4 4 3 4 8 5 1

(a) 2-D latent space



3 1 6 5 7 0 7 6 7 2
8 5 8 4 6 8 2 1 6 2
6 1 5 3 2 8 8 4 3 8
2 8 6 8 4 1 0 0 4 1
5 1 7 2 0 1 5 3 5 4
6 5 6 2 4 4 1 7 8 8
1 3 4 3 9 2 3 7 7 0
4 5 8 2 9 7 0 9 6 4
6 1 4 4 8 7 2 3 4 3
2 6 4 5 6 0 9 7 7 8

(b) 5-D latent space



2 8 7 1 3 0 5 7 3 8
8 3 8 2 7 9 3 3 8
2 5 9 9 4 3 9 5 1 6
1 9 8 8 8 3 3 4 9 7
2 7 3 6 4 2 0 2 0 3
5 9 7 0 5 8 2 8 4 5
6 9 4 3 6 2 8 5 5 7
8 4 9 0 5 0 7 0 6 6
7 4 3 6 2 0 3 6 0 1
2 7 8 0 4 7 7 0 0 0

(c) 10-D latent space



8 2 0 8 7 2 3 7 0 0
7 8 9 9 1 1 7 1 4 4
8 9 6 2 0 3 2 8 2 4
2 9 8 4 3 1 7 0 6 7
5 4 7 1 8 9 7 9 1 0
6 8 2 4 2 4 8 2 8 1
2 5 8 2 4 6 1 3 8 3
7 9 3 9 2 7 9 3 9 6
4 5 2 4 3 9 0 1 5 4
8 8 7 2 5 1 6 2 3 6

(d) 20-D latent space